

*[Instructions: Fill in all details and make sure your supervisor agrees. Delete all **italic** text in and **including** square brackets, including these instructions.*

It should not be longer than about 2 pages. Let your supervisor read and sign. Then submit to thesis coordinator (Cees Diks) to have it signed.

You should complete this form within 2 weeks after the initial indication by the potential supervisor that the topic might be suitable]

Research Proposal

MSc Thesis in Data Science and Business Analytics

Improving Candidate Location Generation in Large-Scale Facility Location Problems Using Road Network and Population Data

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Introduction

Facility location problems seek to find optimal locations for facilities from a set of candidates to best serve a given population. In large-scale applications, the candidate set is typically generated by applying a uniform spatial grid (1, 2, 5 km) over the study region and treating each cell as a possible site. At least two limitations of this approach have been documented in the literature. Theulen (2022) has shown that at 1 km resolution only the feasibility matrix requires approximately 98 GB of memory for the Vietnam case and Slot (2024) observed memory limitations even at 2 km. Although the GRASP heuristic made previously infeasible instances solvable, heuristic complexity still grows with the number of candidates (Slot (2024) observed runtimes of up to 21 hours on the 1 km grid), therefore the underlying problem remains. Secondly, a uniform grid places candidates in uninhabited terrain and areas without road access. Flores et al. (2021) acknowledge this by filtering their 1 km grid down to road-accessible sites only. Their reason for using a grid was computational convenience rather than methodological preference. Together, these studies suggest that grid-based candidate generation is widely used not because it is necessarily optimal, but because it is convenient. An alternative approach could be incorporating road network infrastructure and population distribution as a primary input into candidate location generation, shrinking the candidate set with the aim of reducing computational costs while improving (or maintaining) coverage and solution quality.

Research questions

To what extent does incorporating road network infrastructure and population distribution as primary input into candidate location generation improve solution quality and computational efficiency in facility location problems compared to uniform grid-based approaches?

Methodology and Techniques

Maximum Covering Location Problem (MCLP) will be applied to a generated candidate set derived from two sources: road network intersections extracted from

OpenStreetMap, and gridded population datasets (WorldPop). To measure the effect of this candidate generation approach, solutions obtained using the network and population informed candidate set will be evaluated on coverage achieved and runtime. After that, the results will be compared against existing grid-based solutions reported in the literature on the Vietnam and Timor-Leste datasets.

Data and/or expected results

Population density data (WorldPop) and road network data (OpenStreetMap) are openly available. The expected result is that candidates generated with road network infrastructure and population data as primary consideration will achieve comparable or better coverage than the uniform grid at lower computational cost.

Literature

Theulen, F. (2022). Solving large maximum covering location problems with a GRASP heuristic: Case-study for stroke facility allocation in Vietnam [Master's thesis, Tilburg University].

Slot, A. (2024). Capacitated maximum covering location problem with closest assignment constraints: MIP solver and GRASP heuristic: Case-study for stroke facility allocation in Timor-Leste [Master's thesis, Tilburg University].

Flores, L. J. Y., Tonato, R. R., dela Paz, G. A., & Ulep, V. G. (2021). Optimizing health facility location for universal health care: A case study from the Philippines. PLOS ONE, 16(9), e0256821. <https://doi.org/10.1371/journal.pone.0256821>

Luu, T. T. (2026). Capacity allocation to optimize healthcare accessibility in low- and middle-income countries [MPhil thesis, University of Amsterdam / Tinbergen Institute].

Won, Jun-Jae & Lee, Jong-Seung & Ha, Hyung-Tae. (2025). GIS-Integrated Data Analytics for Optimal Location-and-Routing Problems: The GD-ARISE Pipeline. Mathematics. 13. 3465. 10.3390/math13213465.

Timetable *[of activities and details of intended supervision meetings]*

1st Month

Activity	Date
	1.
	2.

2nd Month

Activity	Date
	3.
	4.

3rd Month

Activity	Date
	5.
	6.

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Supervisor

Signature

Joaquim Gromicho

Second marker (suggestion by supervisor)

Trang Luu

Thesis coordinator
C.G.H. Diks

Signature